Which tag pair should an indie game use? First-Person × Multiplayer

MADT 7202 Mid-Term · Panjapol Ampornratana 6810424009 · Repo: github.com/hhej/steam-tag-whitespace

Steam is the largest PC game store. Players vote tags onto each game; the top tags drive store search.

1 The decision

An indie studio's creative director picks the next game's two lead tags; if no pair passed the selection rule, they would pick a crowded pair (table).

2 Why a network

Counts show how common a pair is and how well it sells, not whether its tags belong to separate groups or which tag connects them. That needs the network.

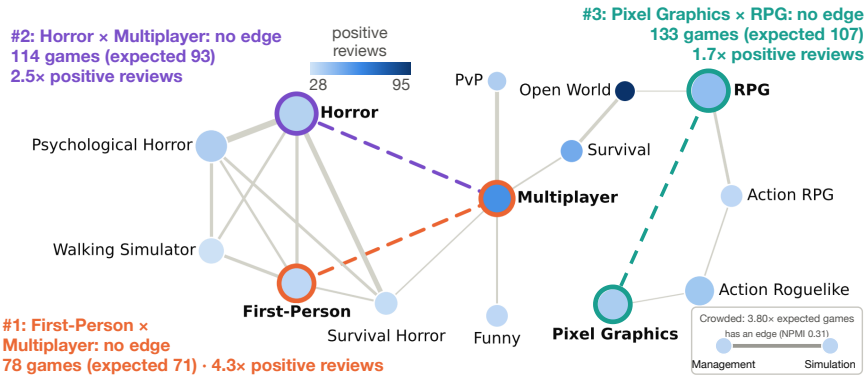
3 Data

Source	S1 FronkonGames steam-games-dataset (Hugging Face); S2 Steam store.
Obtained	S1 downloaded (Sep 2025). S2: 1,000 random S1 games (random seed 7202), scraped 2026-10-14.
Legal	S1 MIT licence; S2 allowed by robots.txt, 1 request/1.5 s; no personal data.
Volume	S1: 9,094 Indie games (2023–24, ≥10 reviews) + 2,076 from Jan–Jun 2025 as a test set. S2: 996.
Check	S1 and S2 agree: 87% top-5 tag overlap; reviews correlate 0.98.
Bias	Favours games that sold; reviews count buyers, not demand; late-2024 games had less time.

4 Graph construction

Node	A Steam tag. Undirected, weighted tag–tag projection of the game–tag bipartite graph.
Edge	Co-occurrence in a game's top 5 tags; suggests relatedness, not a stated relation.
Resolution	Co-op, Online/Local Co-Op and Local Multiplayer merged into Multiplayer (84% of Horror × Multiplayer games were also tagged co-op).
Weight	NPMI (co-occurrence above expected). Raw counts give 9 of the top 10 edges to the 5 most common tags; NPMI gives none.
Window	Top 5 tags per game, as the store shows; with all 20, #1 falls from 4.3× to 2.7×.
Excluded	"Indie", Early Access, Free to Play, adult; on >25% of games: Singleplayer, Adventure, Action, Casual, 2D, 3D, Atmospheric, Exploration.
Pruning	Tag on ≥1.5% of games (136); edge needs NPMI ≥ 0.10 and ≥0.5% of games (45) with both tags. 439 tags → **80 nodes, 177 edges**.

5 The picture



First-Person and Multiplayer have no direct link; the shortest route between them goes through Survival Horror (2 links). Without it, the shortest route is 5 links. 14 of 80 tags; size = games; line width = NPMI.

Expected = games with both tags if picked independently (A's games × B's games ÷ 9,094); *crowding* = games ÷ expected; *positive reviews* = thumbs-up reviews of the pair's typical game. **Rule:** of pairs on ≥73 games, keep crowding ≤ 1.5 and ≤20% of neighbours shared; rank by positive reviews.

Rank: pair	Games	Expected	Crowding	Shared neighbours	Positive reviews	% positive
#1: **First-Person × Multiplayer**	78	71	1.10×	8%	148 (4.3×)	76%
#2: **Horror × Multiplayer**	114	93	1.23×	8%	85 (2.5×)	73%
#3: **Pixel Graphics × RPG**	133	107	1.25×	0%	58 (1.7×)	88%
Crowded (box): **Management × Simulation**	121	32	3.80×	8%	111 (3.3×)	79%
Typical indie game					*34 (1.0×)*	*87%*

6 Findings

- **#1** is on 78 games, near the 71 expected, yet gets 4.3× the typical positive reviews (permutation test, $p < 0.001$). Horror overall gets 1.0×. It wins on reach, not rating: 76% positive.
- **Survival Horror** is the only tag connecting #1's two groups of neighbours: the early warning.
- **Narrow lead:** top at all 8 pruning settings, but in only 60 of 100 resamples. On unseen 2025 games, crowding 1.14 and 2.2× (fewer months to collect reviews).

7 Recommended action

- Week 0: pitch a first-person multiplayer game.
- Weeks 1–6: prototype; playtests must beat 76% positive.
- Week 7: Coming Soon page with both tags. Quarterly: re-run.

8 What would change my mind

If #1's or Survival Horror's crowding passes 1.5, I switch to #2 (1.45 in 2025), then #3. If playtests stay below 76% positive, I switch to #3 (88% positive).